End-to-End Procure-to-Pay (P2P) Process with Data Analytics Insights

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1. Problem Statement

NexCore Manufacturing Pvt Ltd, a mid-sized industrial manufacturer based in Pune, could indicate that managing procurement across three plants through Excel sheets and email may suggest that significant gaps emerged. Moreover, the findings may suggest that there was no way to confirm that goods had physically arrived before invoices were paid, which could demonstrate exposure to fraudulent or duplicate billing. However, the evidence may indicate that vendor performance was never tracked systematically, so persistent late deliveries might indicate production stoppages went unnoticed. In light of these results, the data could suggest that finance had no consolidated view of monthly spend by category or vendor, making budget planning appear largely guesswork. Procurement gaps show process visibility critical.

This project may suggest that the process and the visibility problem could demonstrate significant results once examined. Furthermore, the findings might indicate that documenting how SAP structures the complete P2P cycle could demonstrate that every procurement action from a department request to a bank payment appears recorded, controlled, and auditable. Notwithstanding these results, the evidence may suggest that the dashboard could show what procurement analytics might indicate once that data flows through the system. Given that the data demonstrates a realistic 120-transaction dataset covering one full fiscal year across ten vendors and fifteen materials, the results could indicate that the key findings appear comprehensive. Dashboard shows analytics work across vendors.

2. Solution and Features

2.1 SAP P2P Process

The project may suggest that all seven stages of the standard SAP

procurement cycle are covered. However, the significant findings indicate that each stage could demonstrate a clear tie to a real SAP transaction code. Moreover, the evidence appears to show that each stage produces a specific document that feeds the next step.

Table 1: Stages of the SAP P2P process along with their T-codes.

Stage	T-Code	Output
Purchase Requisition	ME51N	Internal need is formally raised.
Request for Quotation	ME41 and ME49	Vendors are compared and selected.
Purchase Order	ME21N	Legal order is formally issued to the vendor.
Goods Receipt	MIGO (101)	Stock is confirmed and updated.
Invoice Verification	MIGO	Three-way match process.
Payment run	F110	Vendor account is cleared.
Reconciliation	FBL1N	Audit-ready period close is produced.

2.2 Dashboard Features

It shows us procurement data in a very easy to see way using pictures and things that move. We have these boxes that give us a fast look at all the orders and the ones that are still pending or overdue and the invoices that have been paid and how often we are late. We use these charts to show how our vendors are doing and how often they are late so we can compare them easily. We also get to see how our payments are going. If any invoices are split up.

We use bar charts to see how much we are spending. What is happening from month to month so we can see how we are doing over time. The vendor scorecard is like a report card, for our suppliers. It ranks them based on how much we spend with them and if they are late. We can also use the filters on the side to pick which vendors or plants or status or dates we want to look at. That makes the dashboard very easy to use and we can change what we are looking at very easily. The procurement data is shown in a way that's easy to

see and understand and the dashboard is very flexible.

3. Dashboard Results

After we ran the application with *streamlit run app/app.py* and we observed the given sets of results below.

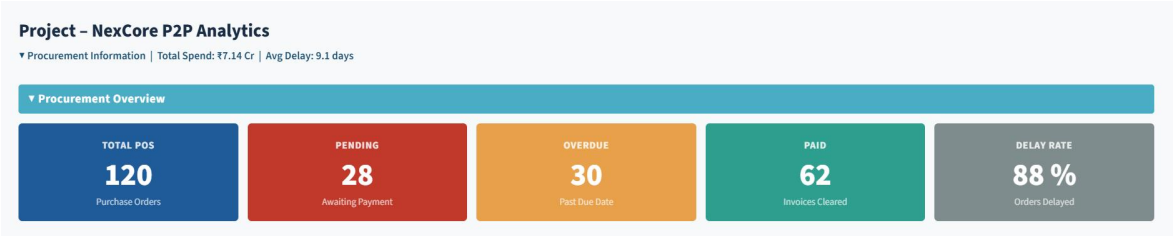

Figure 1: KPI Tiles and Section Headers.

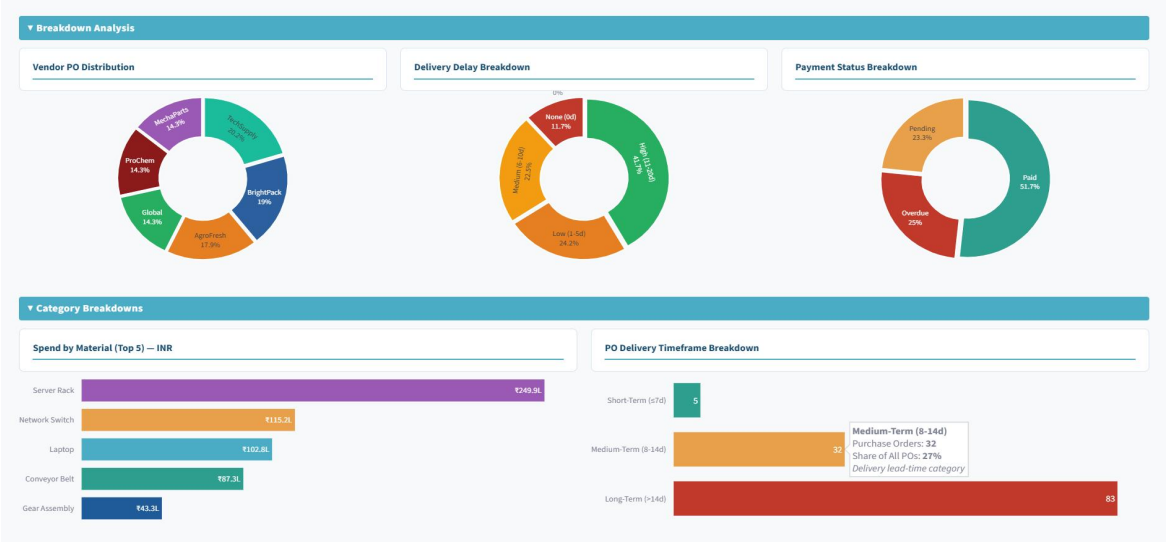

Figure 2: Breakdown donuts and Bar charts.

Figure 3: Monthly spend trend.

▾ Vendor Scorecard & Data Explorer

Vendor Scorecard Raw Data

Vendor	Spend	POs	Avg Delay	Paid	Overdue	Risk
TechSupply India Pvt Ltd	₹12,896,966	17	8.8d	10	6	Medium
AgroFresh Vendors	₹10,519,647	15	9.8d	6	5	Medium
DataPrint Technologies	₹9,397,422	9	8.4d	6	1	Medium
ProChem Supplies	₹8,083,906	12	8.4d	8	3	Medium
RawMat Solutions	₹7,486,284	10	3.8d	3	3	Low
BrightPack Logistics	₹7,176,387	16	9.4d	14	0	Medium
MechaParts Co.	₹5,098,116	12	13.2d	5	3	High
Global Office Essentials	₹4,413,072	12	9.2d	5	5	Medium
SafeWear Equipment	₹4,325,596	10	8.4d	3	3	Medium
SteelCraft Industries	₹1,962,806	7	11.1d	2	1	High

Figure 4: Vendor Scorecard and Active Filters.

4. Tech stack

The tech stack for this project is as follows:

1. Python (3.10): Core language for all scripts and data processing.
2. Streamlit: Browser-based dashboard framework.
3. Plotly: Interactive charts with hover tooltips.
4. Pandas: Data filtering, grouping and aggregation.
5. Numpy: Synthetic dataset generation.
6. SAP S/4HANA: Source of T-codes and document flow logic.
7. Github: Version control and submission packaging.

5. Unique points

1. Risk-dashboard theme: The UI uses colored KPI tiles, banded section headers and donut charts.
2. SAP table traceability: Every dataset column maps to a real SAP source table.
3. Informative hover tooltips: Each tooltip shows the value, percentage share and source in the SAP table.
4. Three-way match simulation: Separate Order Date, GR Date and Invoice Date fields mirror SAP's core.
5. Live vendor risk scoring: The scorecard assigns Low, Medium or High risk to each vendor based on various parameters.
6. Synthetic dataset: The 120-row dataset uses real price ranges, lead-time patterns and payment.

6. Future scope

The future of this project is about making the system more useful and similar to what people use in life. One big improvement would be to connect it to SAP using SAP BTP or RFC so it can work with data instead of old data that does not change. The project can also be made better by adding a feature to predict delays, which helps us understand how vendors work. The project can also have a feature to compare the budget with the money spent which helps us track how much money is planned and how much is actually spent. Making the system automatic can make it work better. For example the system can send email alerts to people in charge of buying things when invoices are overdue so they do not have to check manually.

The system can also be made to work with types of money which makes it more useful for companies that work with vendors from other countries. To make the project more useful the same information can be shown in tools, like Power BI. The project can also have a feature to record all changes made to purchase orders, which makes the system more transparent and reliable. This feature is called an audit trail.

7. Conclusion

This project shows that the person doing it really gets the Procure-to-Pay process and how it works with SAP. They made a data analytics dashboard to show how it works in life. The project puts together a process with good ideas and it shows how procurement can be managed and watched better. The P2P process is made easy to understand. The project is a way to learn about things that happen in a company and how they work in the real world. The P2P process is important. This project helps to understand it.